

Fundamentals of Generative AI & Large Language Models

Concepts · Architectures · Applications · Scaling

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Executive Summary

Generative Artificial Intelligence (Generative AI) is the subfield of machine learning concerned with models that **produce** new data — text, images, code, audio, video, molecules — rather than merely classify or predict it. Over the past decade the field has shifted from narrow, hand-engineered systems to a small number of very large **foundation models** that learn general representations from internet-scale data and can be adapted to thousands of downstream tasks.

The catalyst for this shift was the **Transformer architecture** introduced in 2017. By replacing recurrence with self-attention, Transformers became the first deep-learning architecture that scales almost linearly with available compute and data. When this property is combined with empirical **scaling laws**, model capability improves predictably as parameters, data and compute grow — and a set of **emergent abilities** (multi-step reasoning, in-context learning, tool use) appears once models cross certain size thresholds.

This report provides a structured, technically rigorous overview intended for engineering, product and strategy audiences. It covers the foundational concepts of generative modelling, the Transformer architecture and its modern variants, the application landscape across industries, and the implications — technical, economic and societal — of continued scaling.

Key takeaway. Generative AI is no longer a single technique but a stack: data curation, Transformer-based pre-training, instruction tuning, alignment, retrieval and tooling. Understanding each layer is now a baseline literacy requirement for modern technical leadership.

A Decade of LLM Milestones

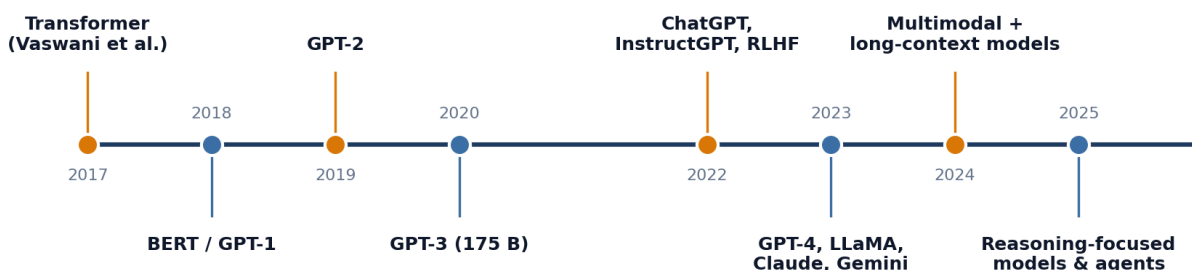


Figure 1 — A decade of milestones in language modelling, from the 2017 Transformer paper to today's reasoning-focused agents.

1. Foundational Concepts of Generative AI

1.1 What Is Generative AI?

At its core, a **generative model** learns the underlying probability distribution $P(\mathbf{x})$ of a data domain — for example the distribution of all natural English sentences, or all human-written Python functions, or all photographs of human faces. Once this distribution is approximated, the model can be sampled from to produce *new* data points that are statistically similar to those it has seen during training but were not part of the training set.

Three properties distinguish modern generative AI from earlier work:

- **Scale.** Models contain billions to trillions of parameters and are trained on trillions of tokens drawn from the open web, books, code repositories and curated datasets.
- **Self-supervision.** Training labels are derived from the data itself — typically by masking or predicting the next token — which removes the need for expensive human annotation at the pre-training stage.
- **Generality.** A single base model, with light adaptation, can perform translation, summarisation, question answering, coding and many other tasks. This is the defining property of **foundation models**.

1.2 Discriminative vs Generative Models

It is useful to contrast generative models with their classical counterparts, **discriminative models**. A discriminative model learns the conditional distribution $P(\mathbf{y} \mid \mathbf{x})$ — given an input $\mathbf{x}$, what is the most likely label $\mathbf{y}$? A generative model learns the joint distribution $P(\mathbf{x}, \mathbf{y})$ or the marginal $P(\mathbf{x})$ directly, and can therefore produce new instances of $\mathbf{x}$.

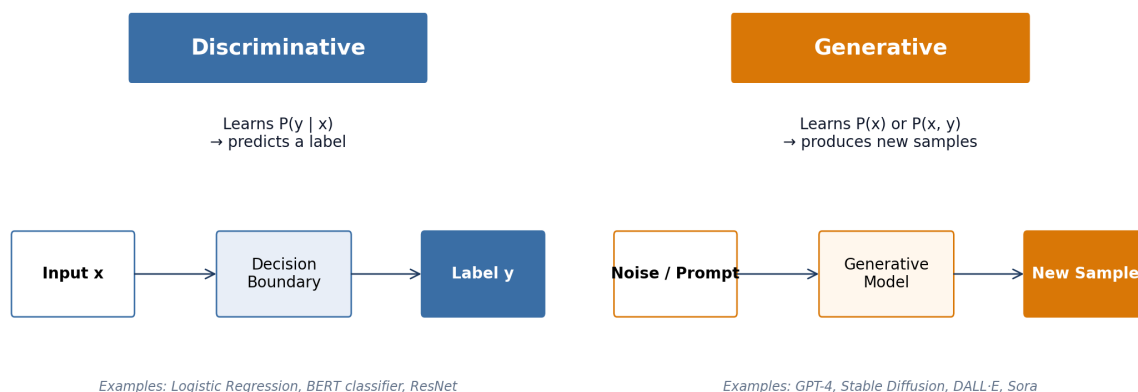


Figure 2 — Discriminative models map inputs to labels; generative models learn the data distribution itself and can sample from it.

1.3 A Taxonomy of Generative Models

Several distinct families of generative architectures have emerged, each with characteristic strengths, training dynamics and failure modes:

Family	Core Idea	Typical Use
Autoregressive (GPT, LLaMA)	Factorise $P(x)$ as a product of $P(x_i x_{<i})$ and predict one token at a time.	Text, code, structured sequences
Variational Autoencoders	Learn a latent distribution $q(z x)$ and decoder $p(x z)$; optimise the ELBO.	Image generation, representation learning
Generative Adversarial Networks	A generator and discriminator play a min-max game until generated samples are indistinguishable from real ones.	Photorealistic images, style transfer
Diffusion Models	Learn to reverse a gradual noising process; denoise pure noise into a data sample.	Image, video, audio, 3-D, molecules
Flow-based Models	Use invertible neural networks for exact likelihood estimation.	Density estimation, scientific data

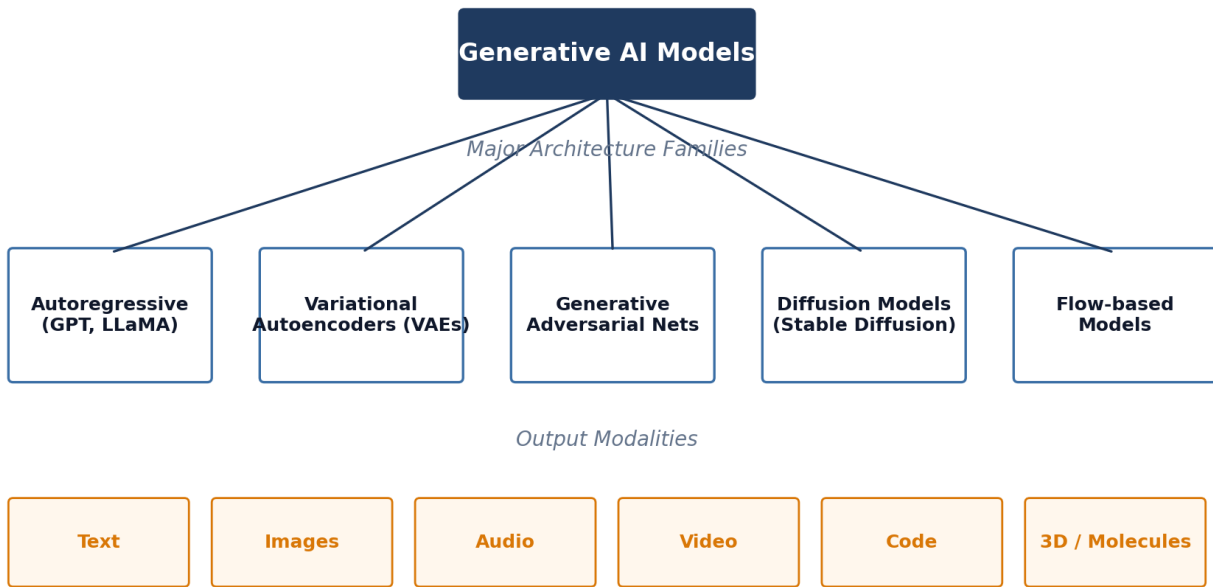


Figure 3 — Taxonomy of major generative model families and the output modalities they support.

Why autoregressive Transformers dominate language. Text is inherently sequential, tokenisation lets us model it discretely, and the next-token prediction objective is dense — every token in a sequence provides a learning signal. Autoregressive Transformers exploit all three properties and have proven, empirically, to scale more reliably than competing approaches for language.

2. Generative AI Architectures

2.1 From RNNs to Transformers

Before 2017, sequence modelling was dominated by **Recurrent Neural Networks (RNNs)** and their gated variants — LSTMs and GRUs. RNNs process tokens one at a time and maintain a hidden state that summarises history. While elegant, this design has two fatal weaknesses at scale: the hidden state becomes an information bottleneck for long sequences, and sequential computation cannot be parallelised across the time dimension, severely limiting training throughput on modern accelerators.

The Transformer, introduced by Vaswani et al. in *Attention Is All You Need* (2017), replaced recurrence with **self-attention** — a mechanism that allows every position in a sequence to look at every other position in a single layer of computation. The result is a model that is both more expressive over long contexts and fully parallel along the sequence dimension, making it almost ideally suited to GPU and TPU hardware.

2.2 The Transformer in Detail

The original Transformer is an **encoder–decoder** network designed for sequence-to-sequence tasks such as machine translation. Its principal components are:

- **Token and positional embeddings** — each input token is mapped to a dense vector; a positional encoding (sinusoidal or learned) injects ordering information, which is otherwise absent from the permutation-invariant attention operation.
- **Multi-head self-attention** — multiple parallel attention operations, each operating in a different learned subspace, allow the model to attend to different kinds of relationships simultaneously (e.g. syntactic and semantic).
- **Position-wise feed-forward networks** — a two-layer MLP applied independently to each token; this is where most of the model's parameters and computation actually live.
- **Residual connections and layer normalisation** — these enable very deep stacks (dozens to hundreds of layers) to train stably.
- **An autoregressive decoder** — the decoder uses *masked* self-attention so that token i can only attend to tokens $< i$, plus cross-attention into the encoder output.

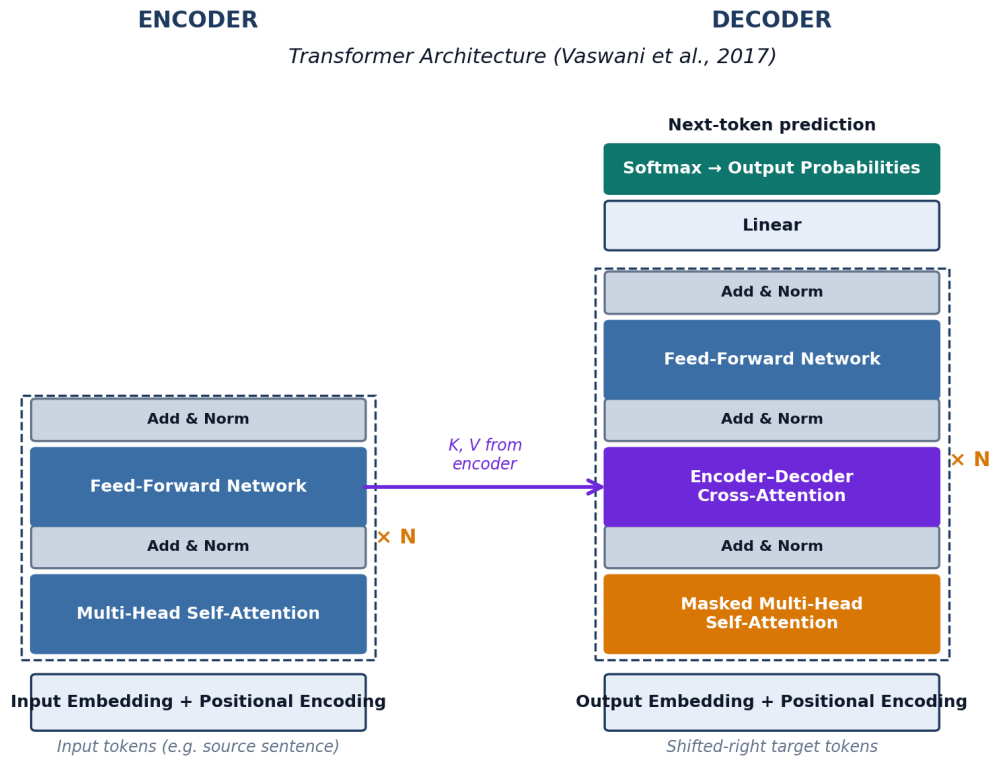


Figure 4 — The original Transformer architecture: an encoder stack (left) and an autoregressive decoder stack (right), connected by cross-attention.

2.3 Self-Attention & Multi-Head Attention

Self-attention is the heart of the Transformer. For each token embedding the model produces three projections — a **query** vector Q , a **key** vector K and a **value** vector V . The output for a given token is a weighted average of all value vectors in the sequence, where the weights are obtained by comparing that token's query against every key:

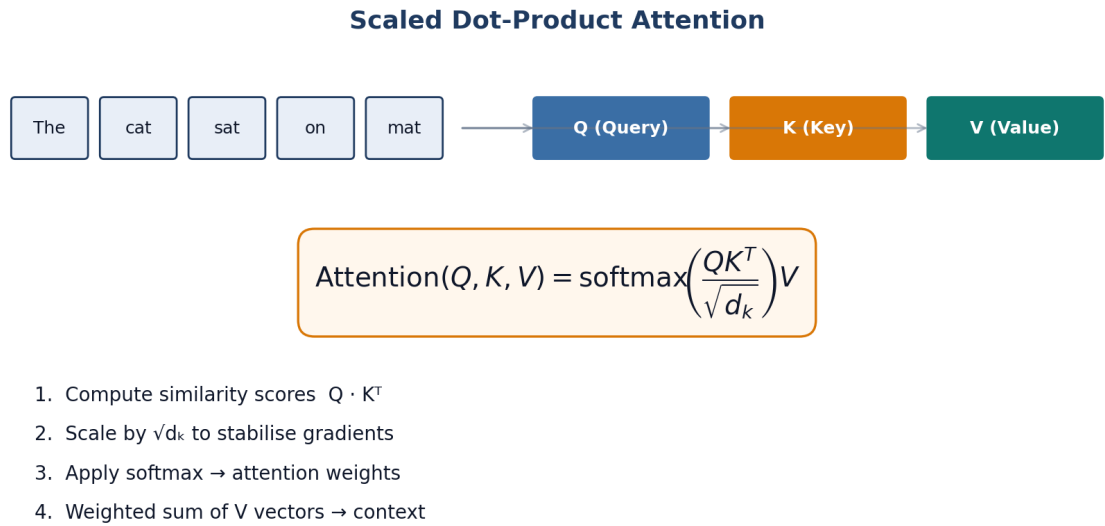


Figure 5 — Scaled dot-product attention. The query, key and value projections enable each token to selectively gather information from every other token in the sequence.

Multi-head attention repeats this operation h times in parallel — typically 8 to 96 heads — each with its own learned Q , K , V projections. The outputs are concatenated and linearly projected. Different heads empirically specialise: some track syntactic dependencies, some attend to coreference, some to long-range topical structure.

Why attention scales. The attention operation has $O(n^2 \cdot d)$ complexity in sequence length n and dimensionality d . Modern long-context models reduce this with sparse attention, sliding-window attention, linear attention or FlashAttention-style memory-efficient kernels — extending usable contexts from 2 K tokens (GPT-2) to 1 M+ tokens (current frontier models).

2.4 Transformer Variants & the Modern LLM Stack

Three Transformer variants now cover most production use cases:

Variant	Examples	Best For
Encoder-only	BERT, RoBERTa, DeBERTa	Classification, embeddings, retrieval, NER
Decoder-only	GPT-3/4, LLaMA, Claude, Gemini	Open-ended text and code generation, chat, agents
Encoder-decoder	T5, BART, Flan-T5	Translation, summarisation, structured transformations

Today’s frontier large language models are decoder-only Transformers trained in a multi-stage pipeline. Pre-training establishes broad knowledge and language competence; supervised fine-tuning teaches instruction following; alignment methods such as **RLHF** (Reinforcement Learning from Human Feedback) and **DPO** (Direct Preference Optimisation) shape the model’s behaviour to match human preferences for helpfulness, honesty and harmlessness.

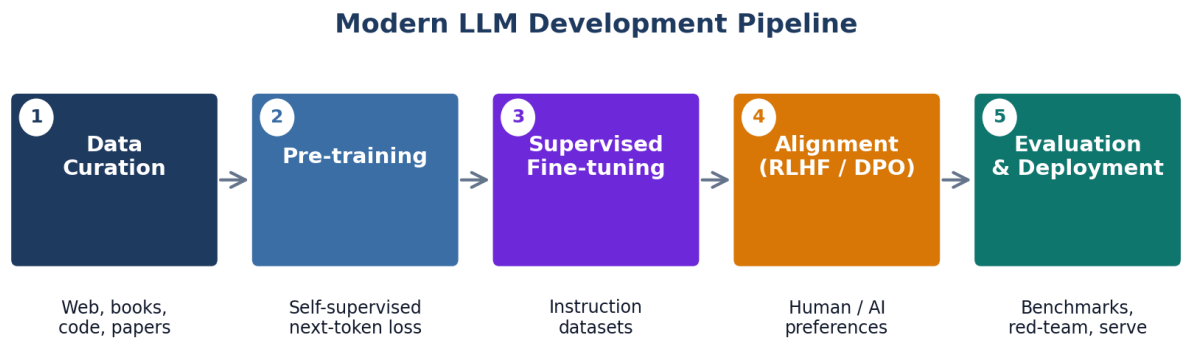


Figure 6 — The modern LLM development pipeline. Each stage uses different data, objectives and feedback signals.

3. Generative AI Applications

3.1 Industry Landscape

Generative AI has moved from research demonstrations to production-grade systems deployed at scale across virtually every knowledge-intensive industry. The unifying pattern is the same: a general-purpose foundation model is grounded in domain context (through retrieval, fine-tuning or tool use) and embedded in an application that augments — and increasingly automates — expert workflows.

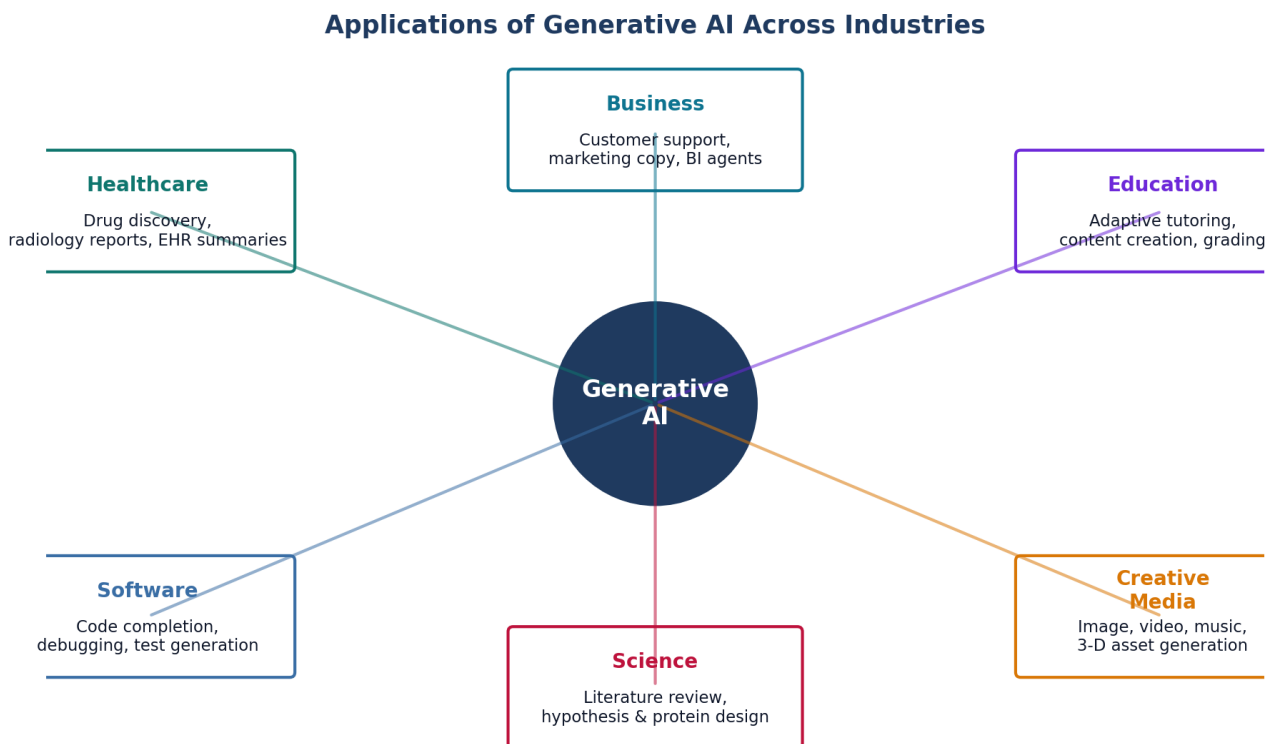


Figure 7 — Generative AI is reshaping six broad industry clusters; the underlying models are largely shared across all of them.

3.2 Concrete Use-Cases by Domain

Software engineering.

Code completion, refactoring, test generation, code review, automated migration between languages or framework versions, and increasingly autonomous coding agents that plan, edit and verify changes across multi-file repositories. This is currently the single most measurable productivity gain from LLMs.

Healthcare and life sciences.

Clinical-note summarisation, radiology report drafting, patient triage chatbots, literature synthesis, and — at the research frontier — protein-structure prediction and de-novo drug design using generative models trained on biomolecular sequences and structures.

Education.

Adaptive tutoring systems that explain concepts at any level of sophistication, automated generation of practice problems and rubrics, accessible language translation, and tools that help teachers personalise material at a scale that was previously impossible.

Creative and media production.

Text-to-image, text-to-video, music composition, voice cloning, 3-D asset generation for games, and an entirely new category of controllable visual editing tools. Diffusion models dominate this category.

Business operations.

Customer-support automation grounded in company knowledge bases, marketing-copy generation, legal-contract review, financial-report drafting, and a rapidly maturing category of **agentic** assistants that can browse, fill forms, query databases and execute multi-step business workflows.

Scientific research.

Literature review and hypothesis generation, materials discovery, weather and climate modelling, and the automation of repetitive experimental analysis. Generative models are increasingly being used as *scientific instruments* rather than just as products.

Adoption pattern. *The most successful enterprise deployments share three traits: (1) the model is grounded in trustworthy data via retrieval-augmented generation; (2) outputs are verifiable by a human or by automated checks; (3) the workflow is bounded — the model assists in a well-scoped task rather than replacing entire roles. Open-ended autonomy remains a research frontier, not a production default.*

4. The Impact of Scaling in Large Language Models

4.1 Scaling Laws & the Compute–Data–Parameters Triangle

Among the most consequential empirical results in modern AI is that the test loss of a Transformer language model decreases as a smooth **power-law** in three quantities — the number of parameters N , the volume of training data D , and the compute budget C — provided the other two are not bottlenecks. Kaplan et al. (2020) characterised this dependency, and Hoffmann et al. (2022) — the *Chinchilla* paper — refined it by showing that, for a fixed compute budget, parameters and tokens should scale **roughly equally**. Earlier models such as GPT-3 were significantly under-trained relative to their parameter count.

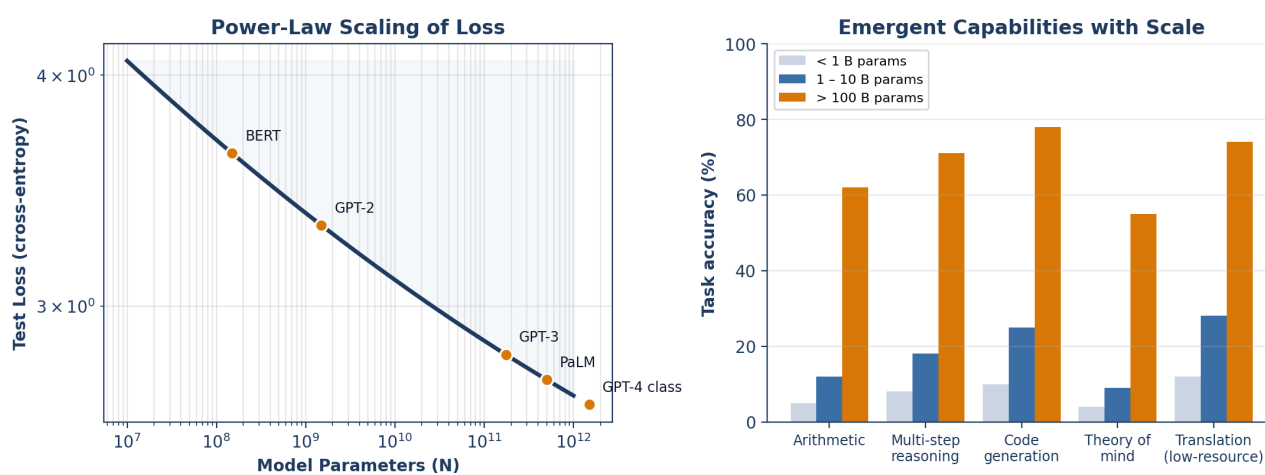


Figure 8 — Left: power-law decrease of test loss with parameter count. Right: a stylised view of how task accuracy increases with scale, with several capabilities only emerging at the largest sizes.

4.2 Emergent Capabilities

While loss decreases smoothly with scale, certain **downstream capabilities** appear abruptly once a model crosses a threshold size. Examples that have been documented across multiple model families include:

- **In-context learning** — solving a novel task from a handful of examples in the prompt, with no gradient updates.
- **Chain-of-thought reasoning** — producing intermediate steps that materially improve performance on multi-step problems.
- **Instruction following** — generalising from instruction-tuning examples to instructions never seen during training.
- **Tool use** — emitting structured calls to external APIs, interpreters or databases and incorporating the results.
- **Multilingual transfer** — solving tasks in low-resource languages by leveraging cross-lingual representations learned at scale.

There is active research debate about whether these abilities are truly discontinuous or whether they appear emergent only because of the metrics used to measure them. Either way, they are reproducible and have been the primary driver of the recent leap in usefulness of LLMs.

4.3 Costs, Risks & Open Questions

Scaling is not free. Training a frontier model now consumes tens of millions to hundreds of millions of dollars in compute, concentrates capability in a small number of well-funded organisations, and carries non-trivial energy and water footprints. Inference cost — long the overlooked side of the equation — is now the dominant economic concern as deployment volumes grow.

Several open technical and societal questions follow from continued scaling:

- **Data exhaustion.** The pool of high-quality public text is finite; future scaling will increasingly depend on synthetic data, multimodal data and proprietary corpora.
- **Alignment and safety.** More capable models are also more capable of producing convincing misinformation, of being misused for cyber-offence, and of behaving in ways their developers did not anticipate. Robust alignment techniques are an open research problem.
- **Evaluation.** Standard benchmarks saturate quickly; evaluating capabilities like reasoning, agency and creativity requires new methodologies.
- **Efficiency.** Architectural innovations (mixture-of-experts, state-space models, retrieval-augmented architectures) and post-training compression (quantisation, distillation) aim to decouple capability from raw scale.
- **Governance.** Concentration of frontier capability in a few actors raises real questions about access, oversight and accountability that are increasingly the subject of regulation in the EU, US, UK and elsewhere.

Strategic implication. *For most organisations, the rational strategy is not to train a frontier model but to compose frontier capability — through APIs, fine-tuning of mid-sized open-weights models, retrieval, evaluation harnesses and good product design. Differentiation will come from data, distribution and workflow integration, not from raw model size.*

5. Conclusion & Outlook

Generative AI has converged, over a remarkably short period, on a simple core recipe: a Transformer-based foundation model, trained with self-supervision on a very large and diverse corpus, then adapted through instruction tuning, alignment and tool use. The **Transformer architecture** made this recipe computationally tractable; **scaling laws** made its capability gains predictable; and a steadily improving set of **alignment** and **retrieval** techniques is making its outputs trustworthy enough to deploy.

Looking forward, several trends are likely to define the next phase of the field:

- **Reasoning-first models** trained with explicit search and verification, narrowing the gap between language fluency and deliberate problem solving.
- **Truly multimodal foundation models** that share a single representational space across text, image, audio, video and action.
- **Agentic systems** that combine planning, memory, tool use and self-verification to complete long, real-world tasks.
- **Smaller, sharper models** obtained through better data, distillation and architectural innovation — making capable AI deployable on-device and at very low cost.
- **Mature evaluation and governance** frameworks that allow organisations and regulators to make principled decisions about deployment.

Generative AI is not the end of the story of machine intelligence — it is, plausibly, only its first chapter. The fundamentals laid out in this report are likely to remain relevant for years even as specific models and benchmarks come and go. A practitioner who deeply understands the data pipeline, the Transformer, the scaling regime and the alignment stack is well-equipped to navigate whatever comes next.

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